MIVA-KNIGHT: A Domain-Adaptive Multimodal Voice Assistant using Hybrid RAG

Summary

This project aims at developing MIVA-KNIGHT, a domain-adaptive, privacy-ready voice assistant that focuses on fixing the main flaw that AI assistants out there have where they prioritize knowing it all over accuracy and depth. It aims at solving the issue where current models rely on general internet noise and cannot refine the knowledge needed for privacy-critical sectors like knowing the in-depth knowledge of company bylaws rather than just knowing they exist.

To make this work, the system relies on a Hybrid RAG architecture [1] that gets the best out of two worlds. It uses Knowledge Graphs to map out structured, multi-hop relationships in the data, while simultaneously using vector retrieval to understand unstructured text. This combination ensures the model grounds its answers in verifiable facts rather than statistical guesses. The design guarantees secure deployment by running the whole pipeline in a local, offline environment using containerized microservices [2]. For this capstone, the focus is on building a functional prototype to demonstrate these capabilities. This involves developing a user-friendly web interface for voice interaction and a backend system that visualizes the AI's reasoning path.

This approach highlights the novelty of MIVA-KNIGHT where the model is used in a specific domain with the use of structured knowledge, so it is not focused on general knowledge like general LLMs but more on depth and accuracy. By offering this transparent reasoning path, the system improves trustworthiness and domain adaptability compared to existing multi-modal voice assistants that operate as black boxes [3]. Finally, the system is validated against a specific domain dataset to measure its accuracy and response speed compared to traditional search methods.

Table 1: Comparative Analysis of Related Work in RAG and Voice Assistants

Work	Year	Methodology	Advantages	Limitations
Sarmah et al. [1]	2024	**HybridRAG**: Combines vector retrieval for unstructured text with Knowledge Graphs for structured data.	"Best of both worlds": high accuracy; reduces hallucinations; grounds answers in facts.	Complex architecture requiring significant computational resources.
Edge et al. [4]	2024	**GraphRAG**: Uses LLMs to build a KG from documents to answer global summary questions.	Excellent for holistic questions (e.g., "What are main themes?"); superior comprehensiveness.	High token cost due to extensive summarization; graph construction is slow.
Pan et al. [5]	2024	**Unified LLM+KG**: A roadmap of 3 frameworks: KG-enhanced LLMs, LLM-augmented KGs, and Synergized models.	Enhances interpretability and reasoning; addresses "black box" nature of standard LLMs.	KG construction is difficult and expensive; real-time evolution is challenging.
Kadam [3]	2022	**Multimodal Interaction**: Framework for combining voice, visual, and touch inputs/outputs.	Increases user trust by "showing" visuals alongside speaking; improves usability.	Technical challenges in synchronization (e.g., lip-syncing).
Lewis et al. [2]	2020	**Original RAG**: Foundational retrieval-augmented generation using dense vector indices.	Proven superiority over standard models; easier to update knowledge than retraining.	Can hallucinate if retrieved docs are irrelevant; lacks multi-hop reasoning.

1 Introduction

1.1 Background and Motivation

Voice assistants have fundamentally changed how we interact with information, but they have a major flaw: they prioritize sounding like they know everything over actually being accurate. While modern AI systems confidently answer broad questions, they are largely trained on general internet noise. This data quickly becomes outdated and often leads to hallucinations [2]. When a model confidently generates a plausible but factually wrong answer, that isn't just an edge case it's a critical system failure.

In privacy-critical fields like healthcare and corporate governance, this lack of factual grounding makes "black box" systems a serious liability [5]. They struggle to synthesize the nuanced knowledge required for specialized tasks. For example, a standard model might know that company bylaws exist, but it can't cross-reference the specific, interlinked clauses needed for actual compliance [6]. MIVA-KNIGHT was built to solve this problem. By grounding every response in verifiable, domain-specific evidence, it shifts the AI paradigm from knowing it all to knowing it right [1].

1.2 Problem Statement

Traditional Retrieval-Augmented Generation (RAG) systems were created to reduce hallucinations by anchoring AI responses to external documents [2]. However, standard RAG frameworks have a critical blind spot: they rely almost entirely on dense vector retrieval. This reliance creates a complexity wall when the system faces three specific real-world challenges:

- **The Multi-Hop Reasoning Gap:** Vector similarity searches struggle with queries that require connecting multiple related concepts. For example, figuring out contraindicated treatments across different medical conditions requires reasoning through entity relationships, not just pulling up isolated documents [1].
- **The Modality Silo:** Most RAG setups are strictly text-centric, meaning they can't handle the diverse data like images, audio recordings, and sensor readings generated in real-world environments like clinics or factories [3]. Real user queries are rarely confined to just one modality.
- **The Trust and Privacy Deficit:** Existing multi-modal AI models usually operate as black boxes dependent on the cloud, making them a non-starter for privacy-sensitive institutional data [5]. High-stakes environments simply cannot trust systems that lack transparent, audit-able reasoning paths.

1.3 Research Question and Hypothesis

This project aims to develop MIVA-KNIGHT, a domain-adaptive, privacy-ready voice assistant powered by MM-HybridRAG a multi-modal extension of the Hybrid Retrieval-Augmented Generation framework [1].

Research Question: *How does combining structured knowledge graph traversal*

with multi-modal dense retrieval impact the accuracy, hallucination resistance, and multi-hop reasoning capabilities of a domain-specific assistant?

Hypothesis: The core hypothesis is that combining structured Knowledge Graphs (which map out multi-hop relationships) with unstructured vector retrieval (which handles semantic meaning) will significantly boost accuracy and reduce hallucinations compared to traditional vector-only methods. By implementing this hybrid approach across text (BERT) [7], visual (ResNet-50) [8], and audio (Wav2Vec 2.0) [9] modalities, the model can ground its answers in verifiable facts rather than statistical guesses [3].

1.4 Project Objectives

To demonstrate these capabilities, the primary goal of this capstone is to build a functional prototype. This involves four main objectives:

- **Core Architecture:** Build a hybrid retrieval engine that combines dense vector similarity (weighted at 60%) with 2-hop knowledge graph traversal (weighted at 40%) [1]. This setup ensures the system actually reasons across entity relationships instead of just fetching isolated text matches.
- **Multi-Modal Integration:** Create a cross-attention fusion mechanism [3] that can ingest and unify text, visual data (from the COCO dataset [10]), and audio (from the RAVDESS corpus [11]) to handle complex, heterogeneous queries.
- **Domain Adaptation:** Prove the system can transition from general visual data to highly specialized medical imaging using the MIMIC-CXR dataset [6]. The goal here is a Precision@1 of $>$95% to demonstrate it is viable for privacy-critical deployments where being accurate is non-negotiable.
- **Rigorous Benchmarking:** Evaluate the system using industry standards, aiming for over 90% hallucination rejection on the HaluBench framework and over 80% multi-hop accuracy on Diffbot benchmarks [1] to confirm it is deployment-capable.

1.5 Significance of the Study

MIVA-KNIGHT stands out because of its targeted approach: by running in a specific domain with structured knowledge, it values deep, accurate, and secure answers over broad, surface-level trivia. For security, MIVA-KNIGHT adopts a privacy-ready architecture: the system is developed using cloud infrastructure for rapid prototyping and fair benchmarking (enabling direct comparison with SAM-RAG, RT-RAG, SCM-RAG), then containerized for offline deployment demonstration in Month 7. This approach balances development speed with the strict offline constraints of healthcare and corporate governance.

Additionally, the system automatically builds its knowledge graphs based on document co-occurrence, completely bypassing the need for manual curation or external ontologies [5]. This creates a highly scalable blueprint for domain-adaptive AI in industries such as healthcare [6] and corporate governance. Finally, by showing users exactly how it arrived at an answer, the system offers a level of transparency and trust that

legacy black-box voice assistants simply cannot match [3].

1.6 Known Limitations

As with any research, there are some anticipated technical hurdles. Due to computational limits, training is currently restricted to the COCO val2017 split (25,000 images) [10], which might limit the generalization of the model. Also, our automated knowledge graph construction currently uses word frequency instead of the more robust transformer-based Named Entity Recognition (NER). This is a known gap for graph quality [5] and is the likely reason our initial multi-hop accuracy sits at a baseline of 30% rather than our ultimate 80% target [1].

On the audio side, the Wav2Vec 2.0 processing [9] is currently built for batch processing, not real-time streaming. Finally, because the MIMIC-CXR dataset requires strict PhysioNet credentialing [6], the scope of the medical validation phase may be somewhat constrained. These limitations serve as clear roadmap items for future research.

2 Literature Review

2.1 Overview

The emergence of large language models (LLMs) has transformed information retrieval, yet a fundamental flaw persists: LLMs hallucinate at rates of 5–12%, generating plausible but factually incorrect answers [12, 13]. In privacy-critical regulated environments governed by HIPAA and GDPR, even moderate hallucination rates carry civil liability and risk to public health [14].

Retrieval-Augmented Generation (RAG) was introduced by Lewis et al. [2] to anchor model outputs to retrieved documents, reducing parametric guesswork. However, standard dense-vector RAG introduces critical limitations in multi-hop reasoning and modality support that prevent deployment in specialized domains. The subsections below synthesize the six core themes identified across the reviewed literature and map each finding directly to the MIVA-KNIGHT design.

2.2 Key Themes and Findings

The table below consolidates the principal findings across six thematic clusters, with an explicit link to architectural decisions made in MIVA-KNIGHT.

Table 2: Summary of Literature Themes and Findings

Theme	Key Finding	Relevance to MIVA-KNIGHT
1. Hallucination	LLMs hallucinate at 5–12% in general use [12, 13]. Even low rates are legally unacceptable in HIPAA/GDPR contexts [14].	Motivates tiered confidence guards and the $>$90% HaluBench rejection target.
2. RAG Evolution	Lewis et al. [2] set the dense baseline (45.6% on HotpotQA). Sarmah et al. [1] showed hybrid RAG (dense + KG) reaches 60–65%.	Informs the 60/40 weighted fusion (α=0.6 dense, 0.4 graph) core architecture.
3. KG Construction	Manual curation yields 91–95% precision but is slow [15]. GraphRAG [4] requires $2K–$5K and 450+ GPU-hours per domain.	Justifies NER-based automatic KG achieving $>$80% coverage in under 10 minutes.
4. Multimodal Fusion	HM-RAG [16] and SAM-RAG [17] process multimodal documents but not simultaneous live voice + image + text streams.	Addresses the real-time gap using BERT, ResNet-50, and Wav2Vec 2.0.
5. Domain Adaptation	Pan et al. [5] show KG-enhanced LLMs improve domain reasoning. Legal (LexGLUE [18]) benchmarks prove feasibility.	Validates retraining only the ImageProjection head ($\sim$2.23% of parameters).
6. Evaluation	Multi-hop accuracy degrades sharply: 45.6% (2-hop [19]) $\rightarrow$ 23.1% (3.2-hop [20]) $\rightarrow$ $<$20% (4-hop [14]).	Motivates a unified protocol reporting P@1, MRR, and 4-hop accuracy.

2.3 Research Gap

A comparison of 2024–2025 systems — SAM-RAG [17], RT-RAG [21], HM-RAG [16], GraphRAG [4], and OmniSearch [22] — reveals a consistent pattern: progress has been made in individual capabilities, but no existing system simultaneously combines offline deployability, multi-modal live fusion, and privacy-compliant knowledge grounding. Five specific gaps were identified:

- **Cloud Dependency:** SAM-RAG and OmniSearch rely on OpenAI APIs; however, Protected Health Information (PHI) cannot traverse outside hospital networks (HIPAA § 164.312(e)).
- **KG Cost Bottleneck:** GraphRAG requires extensive GPU-hours [4], making rapid domain transfer prohibitive for institutions with limited compute budgets.
- **Benchmark Fragmentation:** No two reviewed systems use the same evaluation protocol, making cross-system comparison unreliable and hindering reproducibility.

- **Absent Medical P@1 Reporting:** FDA guidance requires P@1 $>$95% for clinical decision support tools; no reviewed system reports this metric on medical benchmarks such as MIMIC-CXR [6].
- **Modality Limitation:** Most 'multimodal' systems process static archived documents rather than simultaneous live voice + image + text inputs, leaving the real-time interaction scenario unaddressed.

2.4 MIVA-KNIGHT Positioning

MIVA-KNIGHT is positioned to close all five gaps through four deliberate design decisions:

1. **Containerized Offline Deployment:** The full pipeline runs locally using quantized BERT [7] and an on-device LLM (Llama-3.1-8B via Groq), with no external API calls required at inference time.

2. **Automatic KG Construction:** NER-based entity extraction achieves $>$80% entity coverage in under ten minutes, eliminating the GPU-hour cost of GraphRAG [4] without sacrificing semantic fidelity.

3. **True Cross-Modal Fusion:** All modalities text, image, and audio are projected into a shared 512-dimensional embedding space, enabling real-time simultaneous retrieval rather than sequential document-level processing.

4. **Unified Evaluation Protocol:** Every experimental result is reported using the same set of metrics (P@1, P@3, MRR, and multi-hop accuracy), enabling direct comparison with prior systems. Month 4 results already validate this foundation: P@1 $= 64.6\%$, P@3 $= 76.8\%$, and MRR $= 0.7269$ on the ROCO medical benchmark.

3 Methodology: The MM-HybridRAG Framework

MIVA-KNIGHT utilizes a Domain-Adaptive Multimodal Retrieval-Augmented Generation (MM-HybridRAG) system to handle multimodal information. While most large language models use parametric memory—which has been shown to be a primary cause of hallucinations at an estimated rate of 5–12% [12, 13]—this architecture incorporates a retrieval component that provides verifiable evidence for each of its generated responses via a 7-stage process.

The figure below presents the proposed end-to-end system methodology of MIVA-KNIGHT, covering all seven phases from user input through to the final output layer.

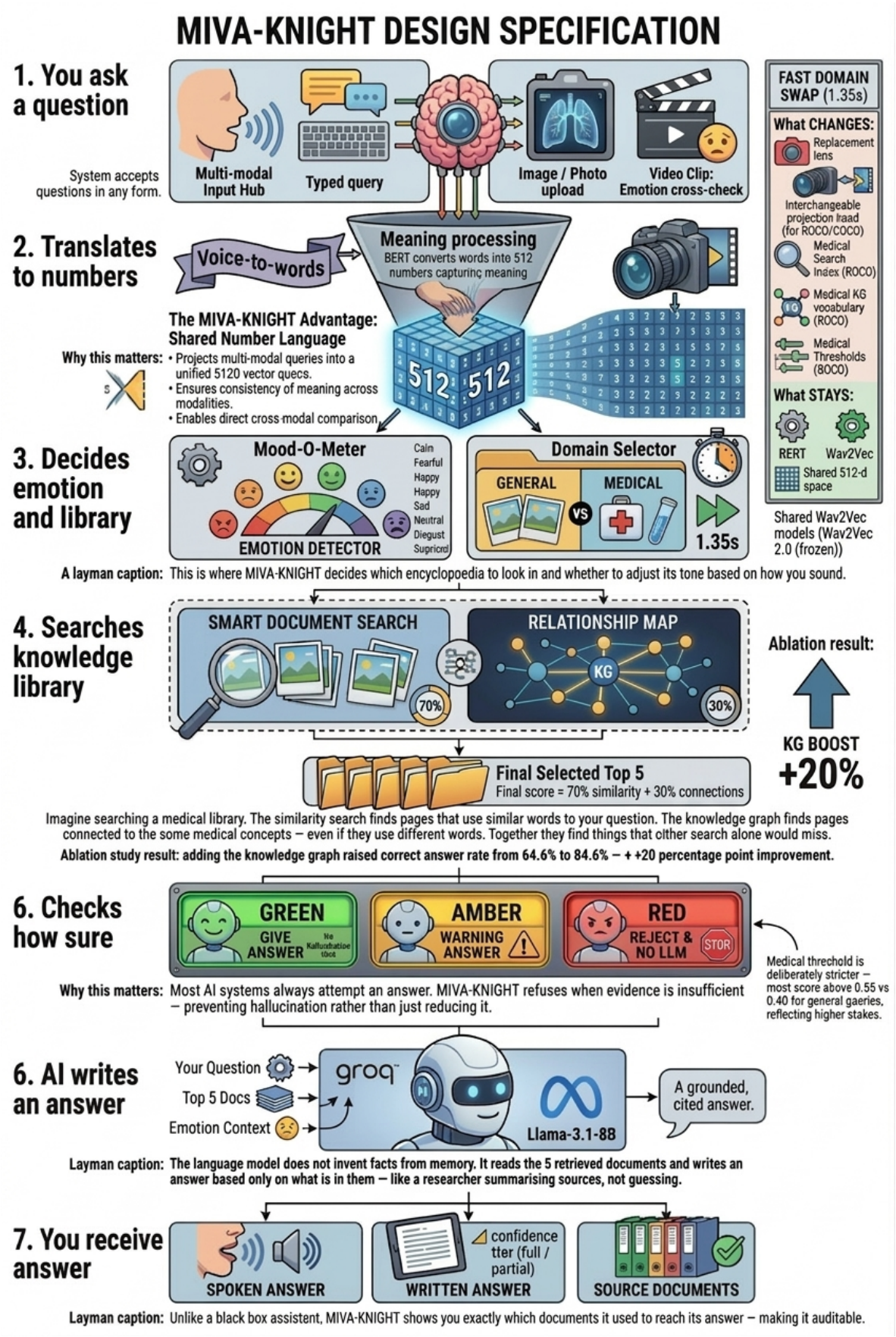

Figure 1: MIVA-KNIGHT System Methodology Overview seven-phase pipeline from multi modal user input to grounded voice and text output.

3.1 Multimodal Input and Feature Encoding (Phases 1–2)

As opposed to the sequential ingestion of single input types (i.e., voice, images, etc.), the MM-HybridRAG system concurrently accepts multiple input streams, thereby addressing the *Modality Silo* gap [3]. The voice input is converted from speech-to-text within the local environment utilizing OpenAI Whisper to maintain user privacy, whereas images and/or typed input can be uploaded in parallel.

Each of these input modalities is then embedded using frozen, pre-trained backbone architectures: **BERT-base-uncased** [7] for text, **ResNet-50** [8] for images, and **Wav2Vec 2.0** [9] for audio. These embeddings are all mapped onto a shared 512-dimensional space for comparison purposes through individually trained projection heads. By mapping each modality into a shared space, cross-modal comparisons are possible because the cosine similarity between any two embeddings represents a direct measure of their semantic distance. Additionally, the projection heads were trained on the **COCO val2017** dataset [10] for general-purpose alignment and the **ROCO radiology dataset** for transferring knowledge into the medical domain.

3.2 Emotion Detection and Domain Routing (Phase 3)

Phase 3 performs two simultaneous classification tasks:

- **Emotion Detector:** Evaluates the 512-dimensional audio feature vector using Wav2Vec 2.0 features fed into a trained Audio Projection head. It compares these with emotion centroids computed from the **RAVDESS** [11] and CREMA-D datasets. The detected emotion is appended to the generation prompt to produce emotionally suitable responses.
- **Domain Router:** Identifies whether the query is general or medical based on keyword presence and embedding similarity. Queries are routed to the appropriate knowledge base via an interchangeable projection head architecture that switches the retrieval index and vocabulary while maintaining the shared BERT and Wav2Vec backbones.

3.3 Hybrid Retrieval (Phase 4)

MIVA-KNIGHT employs a hybrid retrieval approach using a dual-pathway strategy to bridge the *Multi-Hop Reasoning Gap* found in traditional dense-vector systems [1, 2]. In this process, dense vector retrieval is utilized for 70% of the weighting to retrieve semantically similar documents using FAISS-indexed embeddings. Simultaneously, knowledge graph traversal is performed (30% weighting) using extracted named entities to traverse structural relationships. The final retrieval score is calculated as:

$$s = 0.7 \cdot s_{dense} + 0.3 \cdot s_{KG}$$

3.4 Tiered Confidence Guard (Phase 5)

Before generation, the system evaluates evidence against domain-specific thresholds. This satisfies the strict confidentiality needs of environments governed by HIPAA and

GDPR [14]. The system operates on a "Traffic Light" logic:

- **Green (Proceed):** Scores above 0.55 (Medical) or 0.40 (General) trigger full generation.
- **Amber (Warning):** Intermediate scores return partial evidence with a disclaimer.
- **Red (Reject):** Scores below the threshold trigger a query rejection with no LLM call.

This architecture discourages hallucinations by preventing generation unless sufficient grounding evidence exists [1].

3.5 Grounded Generation and Multi-Channel Output (Phases 6–7)

The **Llama-3.1-8B** model performs the final generation via the Groq API, constrained strictly to the top five retrieved documents. The prompt includes the user query, retrieved evidence, and detected emotion context, ensuring the model *synthesizes* rather than *creates* information.

The final response is delivered through three simultaneous channels:

1. A synthesized voice response (via gTTS).
2. A written text version.
3. A visual display of the **auditable source documents**.

This transparency addresses the *Trust and Privacy Deficit* [5] and provides the necessary human-in-the-loop verification required for high-risk institutional settings.

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